Uncovering biological relationships through morphological profiling: The JUMP-Cell Painting dataset and tools

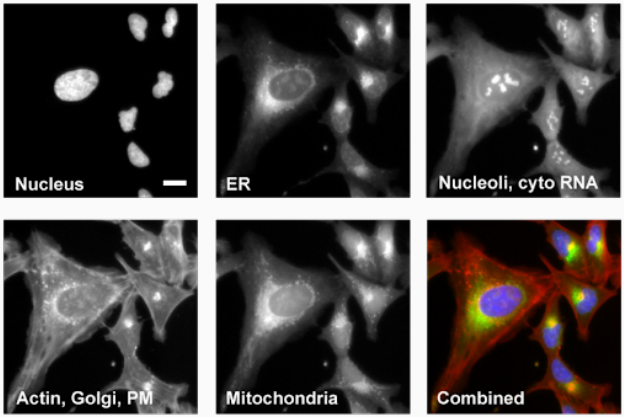
Alán F. Muñoz

Carpenter-Shen Lab, Purdue University; Carpenter-Singh Lab, Broad Institute

10 November 2026

What is Cell Painting?

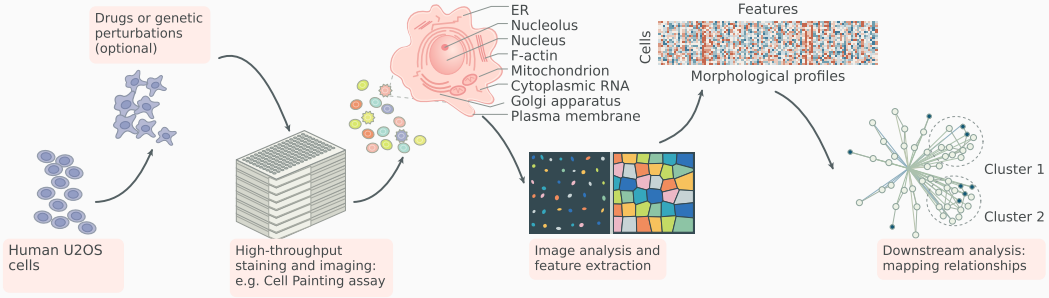
- Fluorescence microscopy of cell morphology.
- Six dyes, five imaging channels.
- Shape, intensity and texture reveal cellular responses.



Source: Muñoz, I2K (2024); assay: Bray et al. (2016)

Cell Painting turns images into measurements

Profiles are numerical descriptions of cells.



Adapted from Chandrasekaran et al. (2021), Fig. 1a

Morphological profiling: How to turn images into profiles

Start with a simple question: are control and perturbed cells different?

Classical features (e.g., CellProfiler)

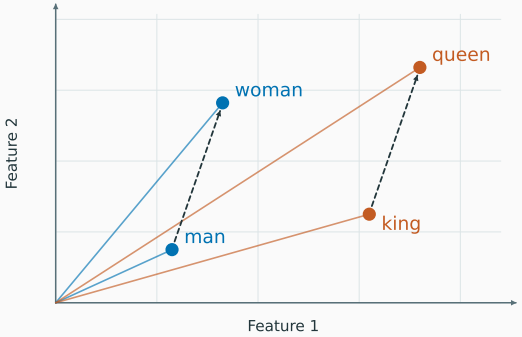
- Size, shape, intensity, texture and relationships between channels and objects.
- Interpretable but partially redundant.

Learned representations

- Features learned from images by neural networks (deep learning).
- Non-redundant numerical vectors (embeddings).

Words as vectors: word2vec

Learning can be viewed as compression



Word → vector

Learned from context

man → woman
king → queen

$$v(\text{king}) - v(\text{man}) + v(\text{woman}) \approx v(\text{queen})$$

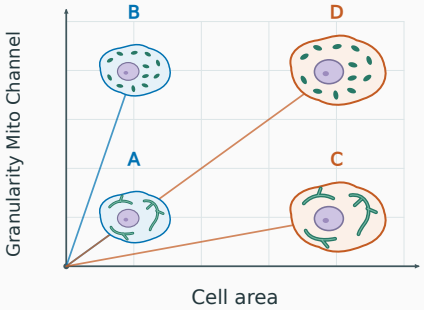
Mikolov et al. (2013)

From words to cells: vectors for comparison

Classical features

Defined measurements

x = [cell area, granularity, ...]

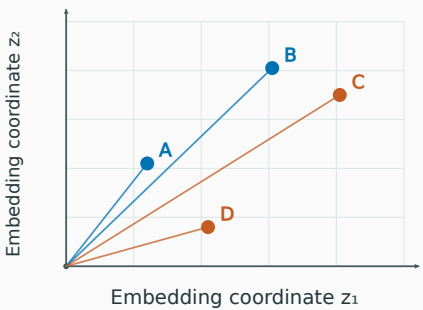


Learned embeddings

Learned from images

Not necessarily directly interpretable.

z = [z_1, z_2, ..., z_n]



Control

Perturbation

JUMP: Joint Undertaking for Morphological Profiling

- Public Cell Painting data: 12 laboratories, around 136,000 perturbations.
- 1.8 billion cells; ~58.3 million images.

CRISPR knockout	Gene overexpression (ORF)	Compound treatment
7,975 gene targets	**12,609** genes	**115,794** compounds

We make JUMP accessible via the JUMP Hub

Choose Your Own Path

1. Just exploring the data?
   - Use a web interface; no programming required. You can look up images for a gene or compound, distinguishing features, most-similar genes or compounds, and more! See the Interactive Tools page for instructions for these tools.
2. Want to fetch and analyze data?
   - Follow our how-to guides for common analyses
3. Looking to build tools?
   - Use our Python libraries cp_measure, cpgdata, jump-portrait, jump-babel and other libraries in the monorepo
   - Access the metadata schema

Biologists can find out:

- Does my gene or compound have a phenotype?
- Which perturbations cause similar phenotypes?
- What changes in cells and organelles?
- Is the phenotype heterogeneous across cells/samples?

Source: JUMP Hub (2026)

Tutorials, how-tos and resources

The Hub contains practical guides to help biologists use JUMP.

How-to guides	No-code gene/compound queries; explore similarity heatmaps.
Tutorials	Curated runnable notebooks for profiles, metadata and statistical analysis.
Further resources	Dataset explanations, protein-localisation analysis and a Guide to Cell Painting for Biologists.

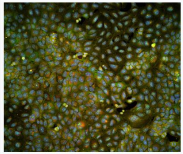
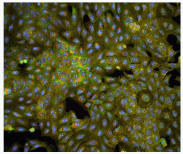
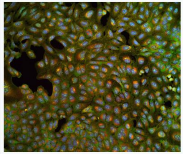
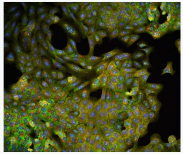
Was my gene or compound tested?

- Find your compound and see whether it was tested.
- Find your gene and see whether it was knocked out or overexpressed.
- View the available profiles, images and metadata.

Link	rowid ▼	JCP2022	NCBI_Gene_ID	standard_key	broad_sample	pert_type
1	1	JCP2022_900002	9	NAT1	ccsbBroad304_00001	trt
2	2	JCP2022_900003	15	AANAT	ccsbBroad304_00002	trt
3	3	JCP2022_900004	18	ABAT	ccsbBroad304_00003	trt
4	4	JCP2022_900005	37	ACADVL	ccsbBroad304_00007	trt
5	5	JCP2022_900006	41	ASIC1	BRDN0001485349	trt
6	6	JCP2022_900006	41	ASIC1	BRDN0001480309	trt
7	7	JCP2022_900006	41	ASIC1	ccsbBroad304_00008	trt

Which perturbations produce related phenotypes?

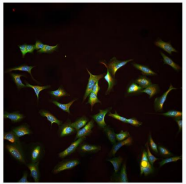
MYT1 example

Gene/Compound	Match	Gene/Compound Example	Match Example	Similarity ▲	Match resources	JCP2022
MYT1	SOS1			0.7195221781730652	External	JCP2022_804400
MYT1	IL13RA2			0.6855948567390442	External	JCP2022_804400

- Precomputed comparisons between perturbation profiles.
- Look for perturbations that match *and* oppose your perturbation of interest.

How does cell morphology change?

AVPR2 overexpression example

Mask	Feature	Channel	Statistic	Gene/Compound	Metadata_image	Median	JCP2022
Cells	Correlation_K_Mito	AGP	8.349928508374835e-29	AVPR2		-0.21309228714749423	JCP2022_905491

- Feature to investigate: mitochondria–AGP overlap.
- Check images: mitochondria versus actin, Golgi and plasma membrane.

Use JUMP as a screen for your phenotype of interest

Feature families

Size / shape	Nuclear area, elongation
Intensity	Mitochondrial staining
Texture	Fine/coarse patterns
Correlation	Between-channel patterns
Distribution	Peripheral/perinuclear signal

Which perturbations change nuclear size?

- Select nuclear area.
- Rank perturbations by feature value.
- Check activity, images and controls.

What do the perturbed cells look like?

MYT1 and PROZ gallery examples

Gene/Compound	External resources	Metadata_JCP2022	Foci 0	Foci 1	Foci 2	Foci 3	Foci 4
PROZ	External	JCP2022_805564					
MYT1	External	JCP2022_804400					
PROZ	External	JCP2022_805564					

Modern tools make image analysis more accessible

Find cells, nuclei and spots	Cellpose (cells and nuclei); Spotiflow (spots)
Follow cells and organelles over time	Trackastra / Ultrack
Measure cell and organelle appearance	cp_measure
Explore your results	marimo

Live demo: ask an agent to query JUMP

Our question

Is SLC2A2 in JUMP? Does perturbing it change cell morphology? Which other perturbations look similar?

Start with documentation, not a guessed answer.

Read the JUMP Hub documentation. Check SLC2A2 availability and reagent IDs. Compare CRISPR and ORF activity separately. Show the source tables and the code you run.

Then ask for evidence.

1. Retrieve similar and opposing profiles.
2. Explain the leading features.
3. Find images and same-plate controls.

External coding agent + public JUMP tools; not a built-in Hub chatbot.

JUMP Hub; worked tutorial

Return with evidence, not just an answer

SLC2A2: a checked rehearsal query, not a new discovery

Perturbation	JUMP reagent ID	Activity corrected p-value
CRISPR	JCP2022_806487	0.00018
ORF	JCP2022_910380	0.09482

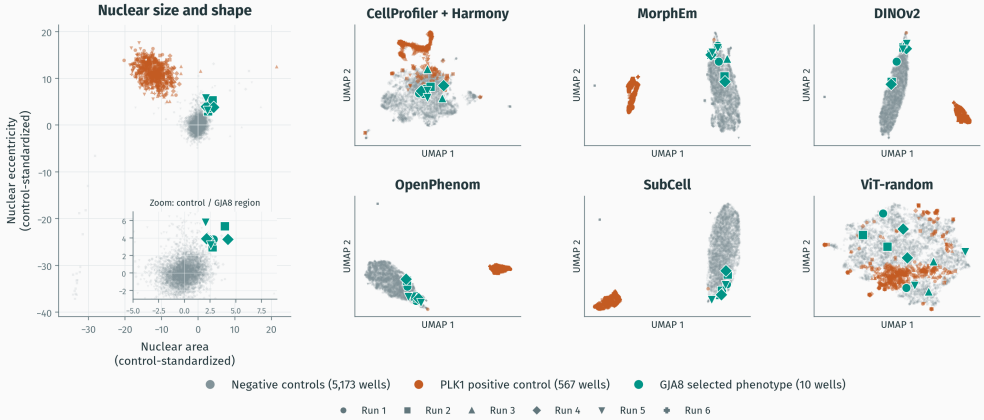
- At 0.05: CRISPR is significant; ORF is not in this analysis.
- Top returned CRISPR match: RAD51 (similarity 0.621).
- Check features, cell counts and images against same-plate controls.

Published ranked subsets, not an exhaustive screen. Similarity suggests hypotheses, not a shared mechanism. Missing significance is not proof of no effect.

Saved tables and code provide an offline fallback; live outputs may change.

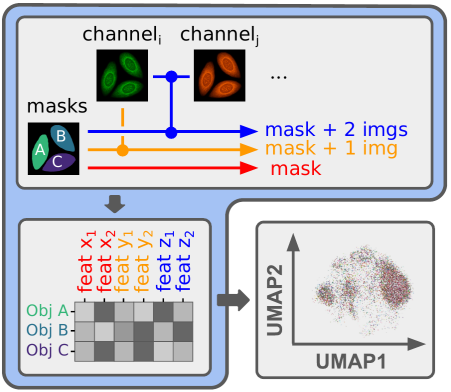
CRISPR features; ORF features; CRISPR matches; queried 9 November 2026

Positive controls contextualize perturbations in both learned and classical representations



Data: JUMP-lite (2026); models: MorphEm, DINOv2, OpenPhenom, SubCell; CellProfiler/Harmony recipe

Not just JUMP: Extract morphological profiles from your own images



cp_measure: feature extraction in Python

- Thousands of features across channels and compartments.
- Fits your Python pipeline.
- Fast software to obtain interpretable morphological features (∼5x CellProfiler).

Adapted from Muñoz et al. (2025), Fig. 1

Applications of JUMP and our tools across biological questions

Evaluate new lysosomal fluorescence probes

Kenneth Li, Vendrell Lab, University of Edinburgh

- Lysosomal acidification probes in cardiomyocytes.
- Uptake, localization and morphology over time.

Compare imaging modalities

Minh Doan, GSK Bioimaging Analytics

- Bright-field time series, Cell Painting, cell cycle and viability.
- Which dyes/imaging methods are better for an different Mechanisms of Action

MisLocus: protein mislocalisation

Johan Fredin, Carpenter-Shen Lab, Purdue University

- Which rare genetic mutations change protein location?
- Compare microscopy of reference and variant proteins.

Integrated in spatial omics software

Tim Treis, Theis Lab, Helmholtz-Munich

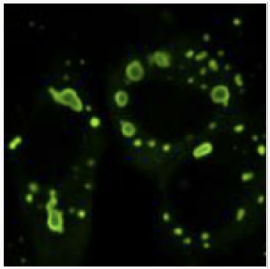
- Integration of morphological features with Squidpy.

MisLocus: a single-cell image benchmark

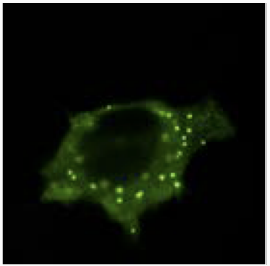
MisLocus contains ∼3 million cells spanning 1,541 reference/variant alleles across 282 genes, excluding controls.

Does a variant change where the protein is located?

LZTR1 reference



LZTR1 G248R



Protein mislocalisation

- Compare reference and variant proteins.
- Test for localisation changes.
- Evaluate image representations.

Unpublished data (preprint coming soon)

Images from Taipale lab

Take-home messages

- JUMP provides public Cell Painting data for around 136,000 chemical and genetic perturbations.
- Biologists can use JUMP Hub to find tested genes and compounds, compare phenotypes and inspect images.
- Open-source image-analysis tools extend morphological profiling to your own experiments.
- Guides and notebooks help scientists or AI agents build analysis workflows.

Thank you

Carpenter-Singh Lab
Anne E. Carpenter
Shantanu Singh

Cimini Lab
Beth Cimini
Paula Llanos

Carpenter-Shen Lab
Anne E. Carpenter
Runxi Shen

JUMP partners



JUMP supporting partners



JUMP collaborators

Scientific:
Horizon Discovery,
a PerkinElmer Company
Administrative:
Pistoia Alliance
Academic:
Umeå University
Stanford Machine Learning Group

Work on cp_measure and JUMP-lite was supported by GSK and NIH MIRA (R35 GM122547).

References

All named tools and models below are clickable.

Assay, data and resources

- Cell Painting (Bray, 2016); profiling review (2021).
- JUMP consortium; counts; centers; images.
- JUMP-lite paper; data/code.
- JUMP Hub; how-to guides; notebooks.
- Guide to Cell Painting.
- Demo recording; Target-2 artifacts.
- word2vec (Mikolov, 2013).

Image-analysis toolbox

- cp_measure; paper.
- CellProfiler; paper.
- Cellpose; StarDist.
- Spotiflow: spot detection.
- Trackastra; Ultrack.
- Nahual: model interfaces.
- marimo; uv.
- Squidpy; Morpheus.

Representations and methods

- DINOv2; DINOv3.
- MorphEm.
- OpenPhenom.
- SubCell.
- ViT-random: DINOv2 code.
- Harmony; JUMP recipe.
- Parquet; Zarr; JPEG XL.

JUMP Hub and slides

JUMP Hub



broad.io/jump

Slides (PDF)



JUMP-lite: a compact benchmark

Same images. Different methods.



Test whether profiles recover known relationships.



Same selected images: 8.7 TB $\rightarrow$ 92 GB ($\sim$94$\times$ smaller; lossy JPEG XL).

Adapted from Muñoz et al. (2026)

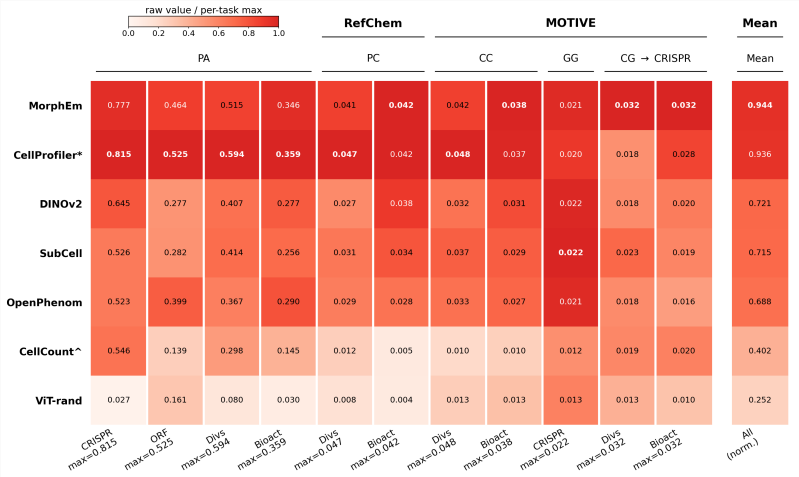
Nahual: swap models within one workflow

Swap models. Keep your workflow.

Image features	DINOv2 ↔ DINOv3 ↔ OpenPhenom ↔ MorphEm
Segmentation	Cellpose ↔ StarDist
Spot detection	Spotiflow
Tracking	Trackastra ↔ Ultrack

JUMP-lite: comparing image representations

Recover replicate perturbations and known biological relationships.

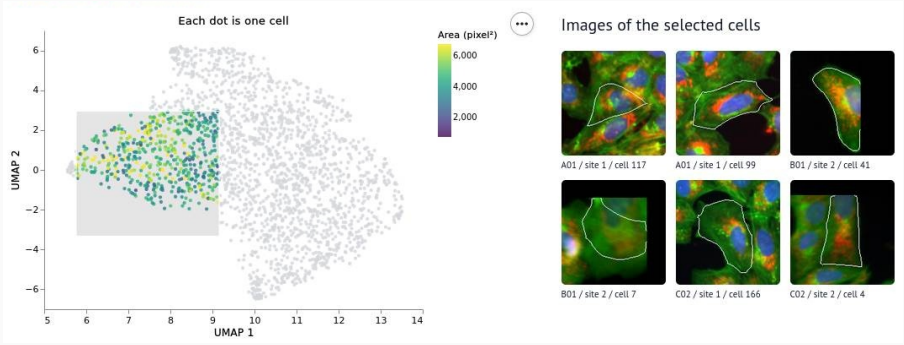


Raw images; colors normalized within each task. CellProfiler/count references use more sites per well.

Adapted from Muñoz et al. (2026), Figure 5

Backup: explore cell subpopulations

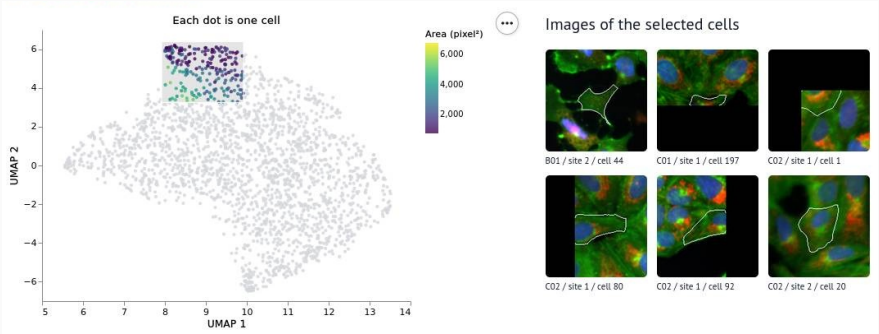
Interactive marimo notebook — separate recorded example



Each dot is a cell: 2,223 from one Target-2 plate. Shown selection: 545 cells. Inspect images, area and other features.

Source: Muñoz (2026), JUMP Target-2 notebook recording

Backup: compare another cell selection



- Second selection: 224 cells; compare images and measurements.
- Not the SLC2A2 query; no treatment/control assignments in this example.

Source: Muñoz (2026), JUMP Target-2 notebook recording